

Week 7 – Skill

Session – 13

Task – 1

Perform input and output redirection in Ubuntu. Create a file, read its contents using input redirection, redirect output to another file, and handle errors using `stderr`

```
ganesh@DESKTOP-79N3772:~$ pwd
/home/ganesh
ganesh@DESKTOP-79N3772:~$
ganesh@DESKTOP-79N3772:~$ mkdir redirection_task
ganesh@DESKTOP-79N3772:~$
ganesh@DESKTOP-79N3772:~$ cd redirection_task
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ echo "Hello Ubuntu" > input.txt
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ cat input.txt
Hello Ubuntu
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ cat < input.txt
Hello Ubuntu
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ cat < input.txt > output.txt
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ cat output.txt
Hello Ubuntu
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ cat output.txt
Hello Ubuntu
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ echo "Linux File Redirection" >> output.txt
ganesh@DESKTOP-79N3772:~/redirection_task$
ganesh@DESKTOP-79N3772:~/redirection_task$ cat output.txt
Hello Ubuntu
Linux File Redirection
ganesh@DESKTOP-79N3772:~/redirection_task$ ls /wrongfolder 2> error.txt
ganesh@DESKTOP-79N3772:~/redirection_task$ cat error.txt
```

Observation

Input, output, and error redirection were successfully performed, and file permissions were modified and verified.

Task 2 — Output Redirection, Files & Permissions

Question:

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, and Test Redirection.

 ganesh@DESKTOP-79N3772: ~

ganesh@DESKTOP-79N3772:~\$ touch output.txt

ganesh@DESKTOP-79N3772:~\$ ls > output.txt

ganesh@DESKTOP-79N3772:~\$ cat output.txt

OS-2520090227

OS-Assignments

OS_Project

Q1_shell

Q2_dispatch

destination.txt

error.txt

filecopy

filecopy.c

fork_exec

fork_exec.c

fork_exec.cy

fork_exec.cy.save

input.txt

linux_io_practical

ls_grep.c

mini_shell

mini_shell.c

os-practical

os_week3

output.txt

process_states

process_states.c

producer_consumer

producer_consumer.c

redirection_task

sample.txt

```
ganesh@DESKTOP-79N3772: ~  
input.txt  
linux_io_practical  
ls_grep.c  
mini_shell  
mini_shell.c  
os-practical  
os_week3  
output.txt  
process_states  
process_states.c  
producer_consumer  
producer_consumer.c  
redirection_task  
sample.txt  
session10  
session11  
session9_shell  
source.txt  
task1_wait  
task1_wait.c  
task1_waitpid  
task1_waitpid.c  
task2_fixed  
task2_fixed.c  
task2_zombie  
task2_zombie.c  
ganesh@DESKTOP-79N3772:~$ date >> output.txt  
ganesh@DESKTOP-79N3772:~$ cat output.txt  
OS-2520090227  
OS-Assignments
```

```
ganesh@DESKTOP-79N3772:~$ ls -l output.txt error.txt result.txt  
-rw-r--r-- 1 ganesh ganesh 60 Sep 29 08:10 error.txt  
-rw-r--r-- 1 ganesh ganesh 537 Sep 29 08:10 output.txt  
-rw-r--r-- 1 ganesh ganesh 74 Sep 29 08:11 result.txt  
ganesh@DESKTOP-79N3772:~$ chmod 644 output.txt
```

Observation:

Output redirection was successfully performed, output and error files were created, permissions were modified and verified, and the redirected data was successfully tested.

Session - 14

TASK 1 — Append Operator and Append Mode

Question:

To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
ganesh@DESKTOP-79N3772:~$ echo "First Line" > data.txt
ganesh@DESKTOP-79N3772:~$ cat data.txt
First Line
ganesh@DESKTOP-79N3772:~$
```

```
ganesh@DESKTOP-79N3772:~$ echo "Second Line" >> data.txt
ganesh@DESKTOP-79N3772:~$ echo "Third Line" >> data.txt
ganesh@DESKTOP-79N3772:~$ cat data.txt
First Line
Second Line
Third Line
ganesh@DESKTOP-79N3772:~$
```

```
ganesh@DESKTOP-79N3772:~$ wc -l data.txt
3 data.txt
ganesh@DESKTOP-79N3772:~$ date >> data.txt
ganesh@DESKTOP-79N3772:~$ cat data.txt
First Line
Second Line
Third Line
Tue Sep 29 08:16:54 UTC 2026
ganesh@DESKTOP-79N3772:~$
```

Observation

Append operator >> successfully added new data while preserving the existing data in the file.

TASK 2 — stderr Redirection

Question:

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

```
Tue Sep 29 08:16:54 UTC 2026
ganesh@DESKTOP-79N3772:~$ ls /nonexistent
ls: cannot access '/nonexistent': No such file or directory
ganesh@DESKTOP-79N3772:~$ ls /nonexistent 2> error.txt
ganesh@DESKTOP-79N3772:~$ cat error.txt
ls: cannot access '/nonexistent': No such file or directory
ganesh@DESKTOP-79N3772:~$ ls /home 2> error.txt
ganesh
ganesh@DESKTOP-79N3772:~$ cat error.txt
ganesh@DESKTOP-79N3772:~$ ls /home /nonexistent 2> errors.txt
/home:
ganesh
ganesh@DESKTOP-79N3772:~$ cat errors.txt
ls: cannot access '/nonexistent': No such file or directory
ganesh@DESKTOP-79N3772:~$
```

Observation

Standard error was successfully redirected to files, and error messages were captured and verified.